

Case Study: Building a Sovereign Cloud Platform for Kenya's HMIS Industry

1. Case Background

The Healthcare Digital Transformation

Kenya's healthcare sector is undergoing a significant transition from fragmented, largely paper-based or disconnected information systems toward an integrated national digital health ecosystem.

The Digital Health Agency (DHA) has been established to develop and operate the infrastructure required for Kenya's comprehensive integrated health information system and national health information exchange.

The regulatory environment has also changed significantly. Under Kenya's Digital Health (Health Information Management Procedures) Regulations, 2025, a healthcare provider or health facility may not use a digital health solution to provide healthcare services unless that solution has been certified by the Digital Health Agency. Certification evaluates functionality, reporting and public-health alerts, information security/privacy/confidentiality, and information exchange/interoperability.

The regulations also place significant requirements on digital health solution providers. A certification application requires evidence including:

- System documentation and requirements specifications
- Data Protection Impact Assessment
- Registration with the Office of the Data Protection Commissioner
- Security, privacy and confidentiality policies
- Backup and recovery policies
- Cybersecurity assessment
- Evidence that the solution can satisfy national interoperability requirements

The national data-exchange framework further requires certified digital health solutions to participate in the national interoperability environment. The regulations establish an enterprise service bus through which certified systems exchange information and establish a Shared Health Record intended to provide a single reference for a patient's medical history.

The HMIS Vendor Problem

This creates a major opportunity—and a major challenge—for Kenya's HMIS vendors.

A typical HMIS company may have a good software product but limited infrastructure expertise. It may currently deploy its software:

- On a hospital's physical server;
- On a small local hosting provider;
- On a public cloud account controlled by the vendor;
- On infrastructure managed by the hospital's IT department; or
- Through a mixture of these approaches.

As the regulatory and technical requirements increase, this model becomes increasingly difficult to scale.

An HMIS vendor serving ten hospitals may suddenly need to address:

- High availability
- Disaster recovery
- Backup
- Cybersecurity
- Encryption
- Access control
- Audit logging
- Data protection
- Business continuity
- Monitoring
- Capacity planning
- Interoperability
- API availability
- Data residency and sovereignty
- Secure connectivity to national health systems

The vendor's core competency, however, is developing healthcare software—not operating data centres.

The Cloud Service Provider

Hynfra Technologies is a Kenyan cloud and infrastructure service provider considering a dedicated offering for HMIS vendors.

The proposed proposition is:

"We provide the secure, sovereign cloud foundation that allows HMIS companies to focus on their software while Hynfra handles the infrastructure, resilience, security and operational complexity."

The service would be hosted in Kenya and could potentially include:

- Compute
- Storage
- Networking
- Managed databases
- Backup
- Disaster recovery
- Cybersecurity
- Monitoring
- High availability
- API hosting
- Disaster recovery testing
- Managed Kubernetes/container platforms
- Secure connectivity
- Infrastructure management

The objective is not to replace the HMIS vendor.

Instead, Hynfra wants to become the **infrastructure and cloud platform behind multiple HMIS vendors.**

2. The Business Opportunity

Imagine that an HMIS company currently has **40 hospital customers.**

Each hospital has different requirements.

Some want the system hosted in the cloud.

Some want dedicated infrastructure.

Some have poor connectivity.

Some require 24/7 support.

Some require a disaster-recovery environment.

Some have strict procurement requirements.

The HMIS vendor currently handles most of this itself.

This creates three problems:

Problem 1 — Cost

The vendor must invest in infrastructure, engineers, monitoring, security tools, backup systems and support.

Problem 2 — Speed

Every new hospital may require a new deployment, configuration and infrastructure design.

Problem 3 — Risk

The HMIS vendor becomes responsible for infrastructure failures in addition to software failures.

A cloud platform that standardizes these requirements could allow the HMIS vendor to onboard new hospitals much faster.

3. The Core Challenge

Your Assignment

You have been hired as a consulting team by Hynfra Technologies.

Your task is to design a **commercially viable sovereign cloud platform specifically for Kenya's HMIS ecosystem**.

The platform must enable HMIS vendors to:

1. Deploy their applications quickly.
2. Meet the infrastructure-related requirements associated with DHA certification.
3. Protect sensitive health information.
4. Provide high availability.
5. Provide reliable backup and disaster recovery.
6. Scale from a small clinic to a large hospital network.
7. Integrate securely with Kenya's national digital health ecosystem.
8. Reduce the HMIS vendor's infrastructure burden.

9. Remain affordable enough for the Kenyan healthcare market.

10. Support rapid onboarding of new hospitals.

Your solution should answer the question:

How can a Kenyan cloud service provider become the preferred infrastructure platform for HMIS vendors while improving their speed to market, regulatory readiness, security and economics?

4. The Key Design Question

Do not assume that "cloud" simply means putting a virtual machine in a data centre.

Consider the complete service.

You need to design:

Infrastructure → Security → Data → Application → Integration → Operations → Commercial Model

The solution should determine what Hynfra provides and what remains the responsibility of the HMIS vendor.

For example:

Responsibility	HMIS Vendor Hynfra	
HMIS application	✓	
Application support	✓	
Application database design	✓	Support
Compute infrastructure		✓
Storage		✓
Network		✓
Backup		✓
Disaster recovery		✓
Cybersecurity	Shared	✓
Monitoring	Shared	✓
Operating systems		✓

Responsibility	HMIS Vendor Hynfra	
Database administration	Shared	✓
DHA certification	✓	Support
Application interoperability	✓	Infrastructure/API support
Data protection	Shared	Shared

Your team should challenge and improve this division of responsibility.

5. Design Constraints

The solution must operate within the realities of the Kenyan market.

Regulatory

The platform must support HMIS vendors in meeting relevant requirements around:

- DHA certification
- Data protection
- Security and confidentiality
- Backup and recovery
- Cybersecurity
- Auditability
- Interoperability
- Data exchange

The students should distinguish between **"the cloud platform is compliant"** and **"the HMIS application is certified."**

Hynfra cannot simply certify an HMIS on behalf of the vendor.

Its role is to provide infrastructure and controls that make the vendor's compliance journey easier.

Data Sovereignty

The proposed solution should prioritize hosting sensitive healthcare information within Kenya.

Students should investigate:

- Where data is stored

- Where backups are stored
- Where disaster-recovery copies are stored
- Who can access the data
- How administrators access production environments
- How encryption keys are managed
- Whether third-party services introduce cross-border data flows

Availability

Hospitals cannot simply stop operating because an HMIS is unavailable.

Students should propose appropriate service levels.

For example:

- Production availability
- Backup frequency
- Recovery Point Objective (RPO)
- Recovery Time Objective (RTO)
- Disaster recovery testing frequency

Economics

The solution must be commercially viable.

Assume that the typical HMIS vendor is price-sensitive.

The students should develop pricing around several customer profiles:

Small clinic

10–20 users

Medium hospital

50–100 users

Large hospital

200–500 users

Hospital network

1,000+ users

The solution should avoid requiring every customer to purchase a large enterprise infrastructure package.

6. Reference Scenario

Hynfra is approached by an HMIS company called **MedSys Kenya**.

MedSys has:

- 35 hospitals
- 80 clinics
- Approximately 3,000 concurrent users
- 150 GB average production database per hospital
- Approximately 500 GB of documents/images per hospital
- 10% annual customer growth
- 24/7 clinical operations
- An existing development team of 15 engineers
- Two infrastructure engineers
- No dedicated security operations team

Its current architecture is inconsistent.

Approximately:

- 40% of customers are on-premise
- 40% are hosted on public cloud
- 20% are hosted through various local providers

MedSys wants to standardize its platform.

It wants to offer hospitals:

"HMIS as a managed service."

It asks Hynfra to design the infrastructure platform.

7. Technical Requirements

The proposed platform should consider:

Application

- Virtual machines and/or containers
- Application load balancing
- Separate production and non-production environments
- Development/test environments
- Automated deployment
- Application isolation between customers

Database

Consider:

- Database clustering
- Replication
- Encryption
- Automated backups
- Point-in-time recovery
- Database monitoring

Storage

The solution should distinguish between:

- Operating system storage
- Database storage
- Application storage
- Patient documents
- Images
- Backup storage
- Archive storage

Networking

Consider:

- Internet connectivity
- Hospital-to-cloud VPN

- Private connectivity
- Firewalling
- Network segmentation
- API connectivity
- Administrative access

Security

Consider:

- Identity and access management
- MFA
- Encryption at rest
- Encryption in transit
- Privileged access
- Endpoint security
- Vulnerability management
- SIEM/SOC
- Intrusion detection
- Audit logs

Backup

Design:

- Backup frequency
- Retention
- Immutable backups
- Offsite copies
- Backup testing
- Recovery testing

Disaster Recovery

Design at least two scenarios:

Scenario A — Infrastructure failure

A production server fails.

Scenario B — Site failure

The primary data centre becomes unavailable.

Students should specify the proposed RTO and RPO for each scenario and explain the economics of achieving them.

8. Interoperability Challenge

The HMIS must participate in Kenya's national digital health ecosystem.

The data-exchange regulations establish an enterprise service bus for routing messages between certified digital health solutions and establish national shared resources and a Shared Health Record. Certified digital health solutions must query/update the Shared Health Record for encounters and maintain auditable access logs.

Students should therefore design an **integration architecture** showing:

Hospital → HMIS → Hynfra Platform → National Digital Health Infrastructure

The design should consider:

- APIs
- API gateway
- Authentication
- Encryption
- Message queues
- Logging
- Monitoring
- Retry mechanisms
- Data transformation
- Error handling
- API availability
- Disaster recovery

Students should explicitly identify which components belong to Hynfra and which belong to the HMIS vendor.

9. Commercial Challenge

Technology alone is not enough.

Hynfra needs a business model.

Students should design a pricing model that could include:

Option A — Per hospital

A fixed monthly infrastructure fee.

Option B — Per user

Infrastructure priced according to the number of users.

Option C — Per workload

Compute, storage, backup and network charged separately.

Option D — Platform subscription

The HMIS vendor pays Hynfra a wholesale platform fee and resells the service to hospitals.

Option E — Hybrid

A base platform fee plus consumption-based charges.

Students should determine which model best aligns Hynfra's interests with those of HMIS vendors.

10. Competitive Analysis

Students should compare Hynfra's proposition against:

- AWS
- Microsoft Azure
- Google Cloud
- Local Kenyan cloud providers
- Hospital-owned infrastructure
- Traditional data-centre hosting
- VMware-based private clouds
- Other sovereign cloud providers

The analysis should consider:

Factor	Hynfra Hyperscaler On-Premise
---------------	--------------------------------------

Data sovereignty

Cost

Scalability

Local support

Disaster recovery

Security

Availability

Compliance support

Deployment speed

Technical expertise required by HMIS vendor

11. Expected Deliverables

The student team should produce the following.

Deliverable 1 — Industry Analysis

A 5–10 page analysis covering:

- Kenyan healthcare digitization
- HMIS market
- DHA requirements
- Key challenges facing HMIS vendors
- Cloud adoption barriers
- Competitive landscape
- Market opportunity

Deliverable 2 — Target Customer Profile

Define the ideal HMIS customer for Hynfra.

Include:

- Company size
- Number of hospitals
- Technology maturity
- Hosting model
- Pain points
- Buying triggers
- Decision-makers
- Budget considerations

Deliverable 3 — Reference Architecture

Produce a detailed architecture diagram showing:

Hospital → Connectivity → Security → Hynfra Cloud → HMIS Application → Database → Backup/DR → National Digital Health Integration

Include:

- Network zones
- Security controls
- Compute
- Storage
- Database
- Backup
- DR
- Monitoring
- API integration
- Management plane

Deliverable 4 — Security & Compliance Framework

Map the proposed architecture to:

- DHA certification requirements
- Data protection requirements
- Backup/recovery requirements

- Cybersecurity
- Access control
- Encryption
- Audit logging
- Data sovereignty

The team should identify which controls are:

Hynfra responsibility / HMIS vendor responsibility / Shared responsibility

Deliverable 5 — Disaster Recovery Design

Define:

- RPO
- RTO
- Backup architecture
- DR architecture
- Failover process
- Recovery testing
- DR testing frequency

Provide at least two service tiers.

Deliverable 6 — Commercial Model

Design a three-tier offering.

For example:

Essential

Small clinics

Professional

Medium hospitals

Enterprise

Hospital groups

For each tier specify:

- Compute

- Storage
- Backup
- DR
- Security
- Support
- SLA
- Price

Students should show their assumptions and unit economics.

Deliverable 7 — Go-to-Market Strategy

Design Hynfra's strategy for acquiring its first 20 HMIS customers.

Include:

- Ideal customer profile
- Lead-generation channels
- Value proposition
- Sales pitch
- Partnerships
- Pricing strategy
- Proof-of-concept strategy
- Customer onboarding process

Deliverable 8 — Business Case

Produce a five-year financial model covering:

- Infrastructure investment
- Operating costs
- Revenue
- Gross margin
- Customer acquisition
- Customer growth
- Break-even

- ROI

Students should identify the assumptions that have the greatest impact on profitability.

Deliverable 9 — Executive Pitch

Create a **10-slide board/investor presentation** answering:

Why should Hynfra become the cloud platform of choice for Kenya's HMIS vendors?

The presentation should demonstrate that the opportunity is technically feasible, commercially attractive and defensible.

12. Evaluation Criteria

Teams will be assessed on:

Criterion	Weight
Understanding of healthcare/HMIS problem	15%
Regulatory & compliance understanding	15%
Technical architecture	20%
Security & resilience	15%
Commercial model	15%
Go-to-market strategy	10%
Quality of assumptions/data	5%
Presentation & clarity	5%
Total	100%

13. The Twist

Halfway through the project, introduce the following scenario:

MedSys has just won a contract to deploy its HMIS across 25 additional hospitals. The contract requires the system to be operational within 90 days.

The hospitals have different connectivity providers, different legacy systems and varying levels of IT maturity.

MedSys does not want to hire additional infrastructure engineers.

The hospitals require backup, disaster recovery, cybersecurity and 24/7 availability.

The HMIS vendor expects Hynfra to provide a standardized platform that can onboard each hospital rapidly.

The teams must revise their architecture and commercial proposal.

Final question

Can Hynfra turn infrastructure from an obstacle for HMIS vendors into a competitive advantage—allowing them to deploy faster, satisfy increasingly demanding regulatory requirements, and scale across Kenya without building their own cloud infrastructure?